

# Python Programming

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## Unit 1: Introduction

Installing Python, simple program using Python, Expressions and values, variable and computer, Memory, error detection, multiple line statements, designing and using functions, function provided by Python, Tracing function calls in memory model, omitting return statement, working with text: creating string of character, using special character in strings, creating a multiline string, Printing Information, getting information from the keyboard, A Boolean type, choosing statements to execute, Nested If Statements, remembering the results of a Boolean expression Evaluation

## Unit 2: A Modular Approach to Program Organization

Importing Modules, Refining Your Own Modules, testing Code Semi automatically Grouping Functions using Methods: Modules, classes and Methods, calling Methods the object oriented way, exploring string methods, underscores, storing collections of data using lists: storing and accessing data in lists, modifying lists, operations on lists, slicing lists, aliasing, List methods working with a list of Lists.

## Unit 3: Repeating code using Loops

Processing items in a list Processing characters in strings, looping, over a range of Numbers, Processing lists using indices, Nesting loops in loops, looping until a condition is reached, Repetition based on user input, controlling loops using Break and continue. Reading and Writing Files: kinds of files, opening a file, Techniques for Reading Files, Files over the Internet, writing Files and Writing Algorithms that uses the File reading techniques, Multiline Record.

Unit 4: Storing data using other collection Types  
Storing data using sets, storing data using  
tuples, storing data using dictionaries, Inverting  
a Dictionary, using the In operator on Tuples,  
sets and dictionaries, comparing collections, collecting  
New information, object oriented Programming:  
Understanding a Problem Domain, Function  
Instances, class object and class book, writing  
a method in class Book

Plugging into Python Syntax: More special  
methods, Creating Graphical user Interface (GUI),  
Building a basic GUI, Models, views and  
controllers, customizing the visual style, introducing  
few more widgets, object oriented GUIs, keeping  
the concepts from being a GUI mess.